

Bayes Factor
Power

Tilburg
October 10th

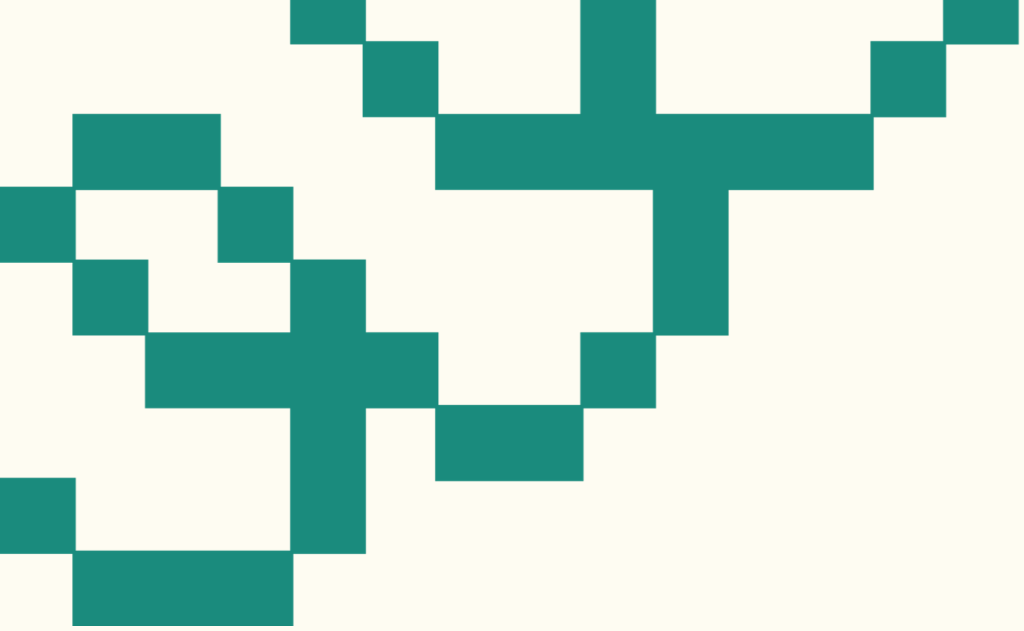


Utrecht
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Sharing science,
shaping tomorrow

BAYES FACTOR POWER: SAMPLE SIZE DETERMINATION IN CLUSTER RANDOMIZED TRIALS

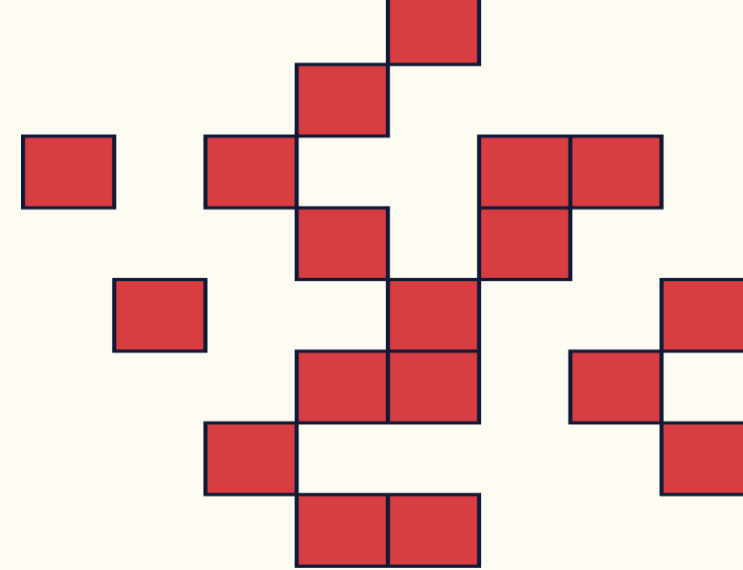




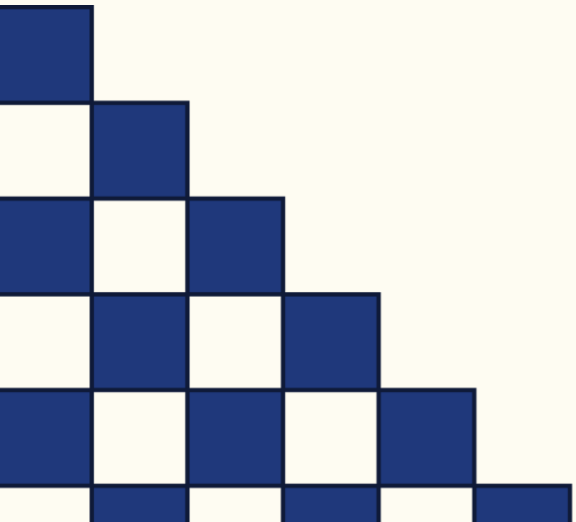
CONTENT

1. Introduction
2. Theoretical Background
 - a. Bayesian Power
 - b. Bayes Factor
 - c. Cluster Randomized Trials
3. BayesSSD
4. Applications

WHY AM I HERE?



Lack of free software for Bayesian sample size determination for multilevel models.



FREQUENTIST POWER

$$\text{Power} = P(p < \alpha | H_0 \text{ false})$$

Evidence can exclusively be obtained against the null

	H_0 is true	H_0 is false
H_0 not rejected		Type-II error (β)
H_0 rejected	Type-I error	

$$\text{Power} = 1 - P(\text{Type-II error}) = 1 - \beta$$

BAYESIAN POWER

- Evaluate a pair of competing hypotheses (Bayes factor)
- Find the best hypothesis out of a set (Posterior Model Probabilities)

BAYESIAN POWER

- Probability of finding a convincing Bayes Factor (BF) in favour of the *true* hypothesis

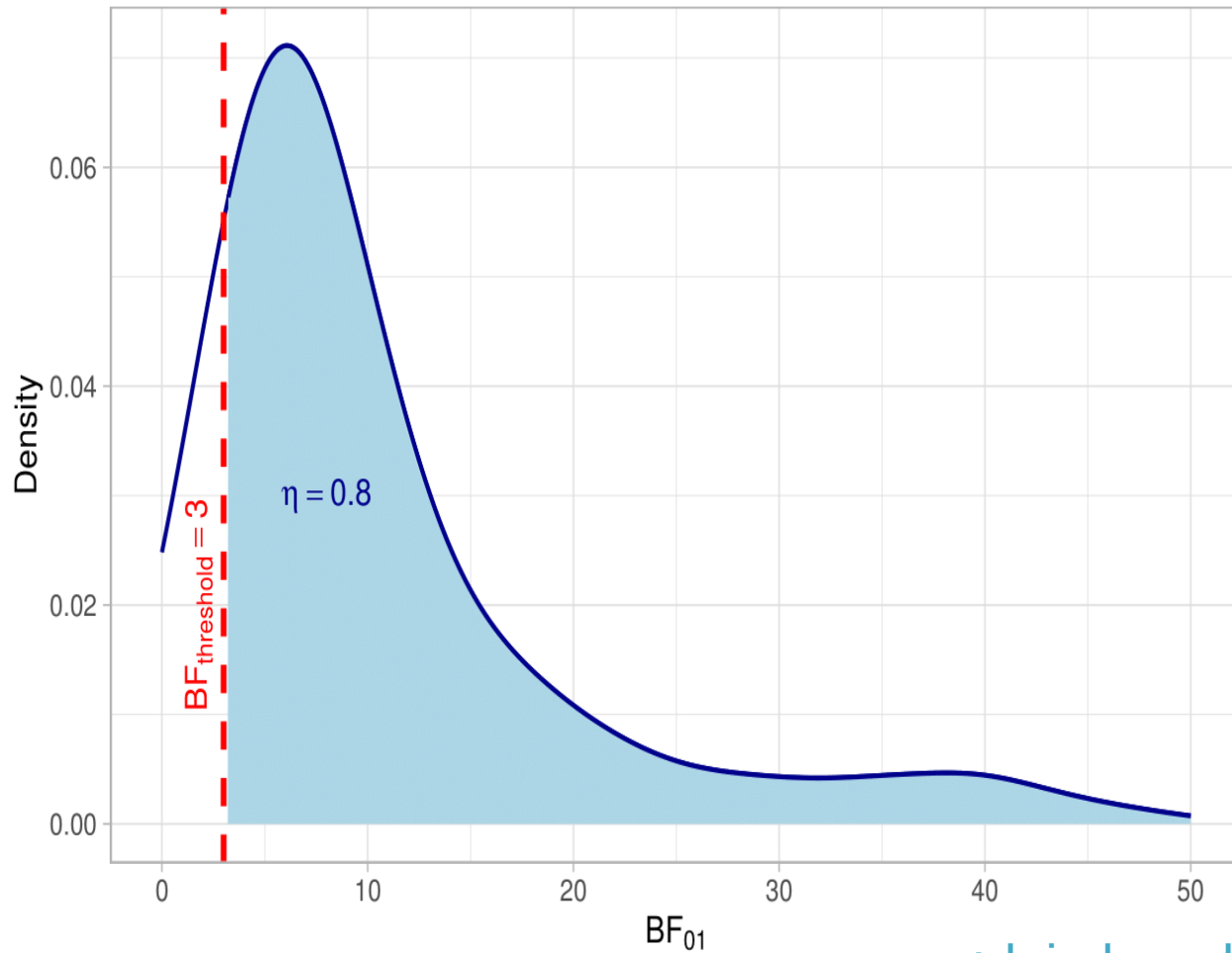
$$P(BF_{10} > BF_{thres} | H_1) \geq \eta$$

	H is not the best	H is the best
H is not selected		Type-II error (β)
H is selected	Type-I error	

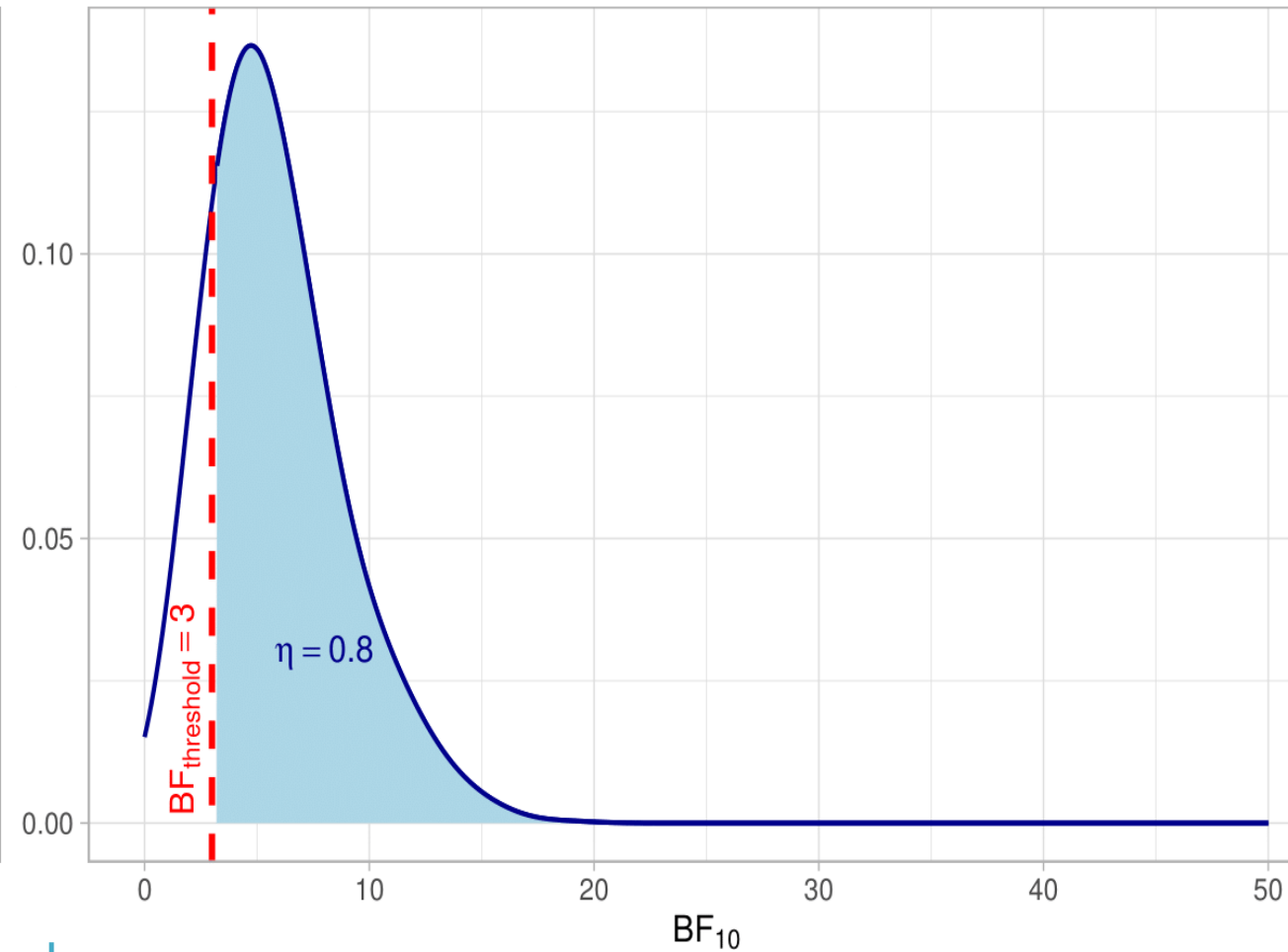
$$\text{Power} = 1 - P(\text{Type-II error}) = 1 - \beta$$

BAYESIAN POWER

$H_0 \neq \text{true}$



$H_1 \neq \text{true}$



η : desired power level

BAYESIAN HYPOTHESIS TESTING

Quantifies relative evidence for a pair of competing hypotheses

$$BF_{10} = \frac{Hypothesis_1}{Hypothesis_0}$$

- > 1: Hypothesis 1 is preferred
- = 1: None of the hypotheses is preferred
- < 1: Null hypothesis is preferred

Hypothesis 1 has x times larger support than hypothesis 2

The support for hypothesis 1 is x times larger than for hypothesis 2

Quantification of the relative support of the data in a scale between 0 - 1

$$PMP_{H_1} = \frac{BF_{1u}}{\sum_{m=1}^M BF_{mu}}$$



We assume that the prior probabilities for each hypothesis are equal.

Bayes
Factor (BF)

Posterior
Model
Probabilities
(PMP)

BAYESIAN HYPOTHESIS TESTING

Approximated

Prior and posterior distributions are normally approximated where the mean is the maximum likelihood

$$\pi_u(\theta|X) \approx N\left(\theta, \frac{\widehat{\Sigma}_\theta}{b}\right)$$

Adjusted

The prior distribution is centred around the focal point of interest

$$\pi_u^*(\theta|X^b) = N\left(\theta^*, \frac{\widehat{\Sigma}_\theta}{b}\right)$$

Fractional

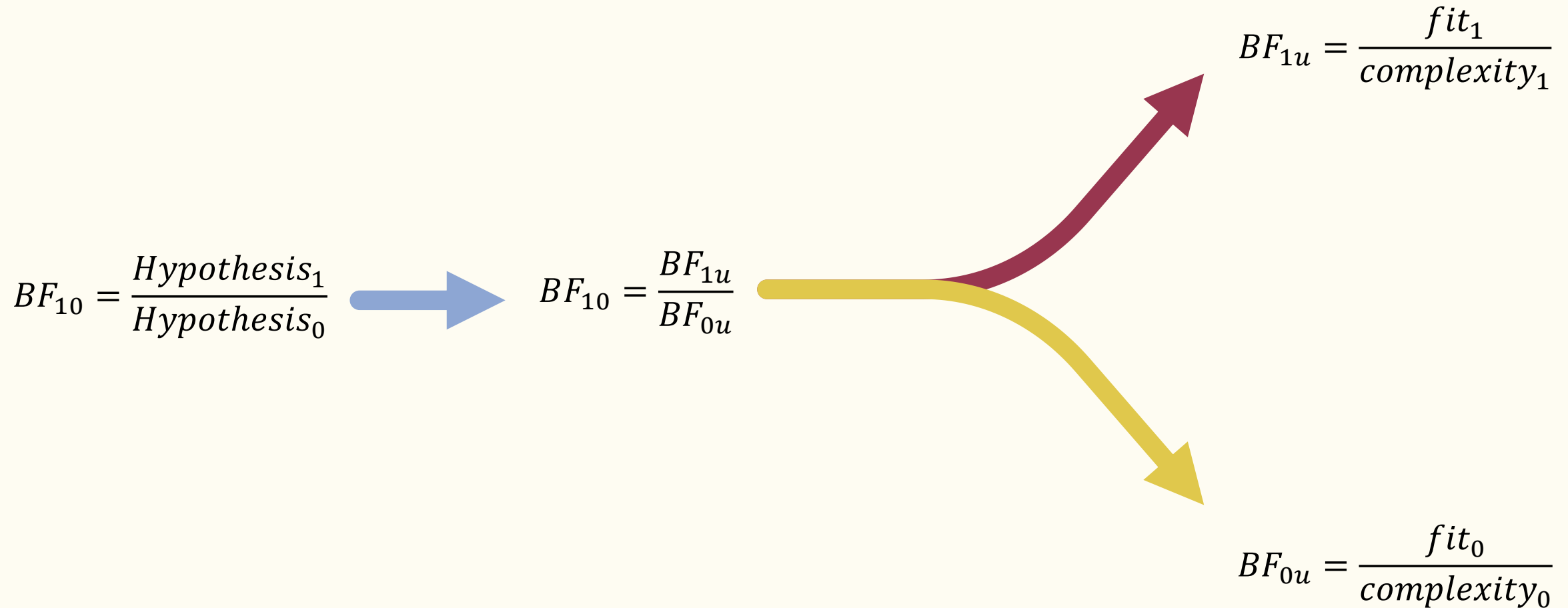
A fraction of information (*fraction b*) in the data is used to specify the prior

$$\pi_u^*(\theta|X^b) = N\left(\theta^*, \frac{\widehat{\Sigma}_\theta}{b}\right)$$

Bayes Factor

$$AAFBF_{iu} = \frac{\int_{\theta \in \Theta_i} \pi_u(\theta|X) d\theta}{\int_{\theta \in \Theta_i} \pi_u^*(\theta|X^b) d\theta}$$

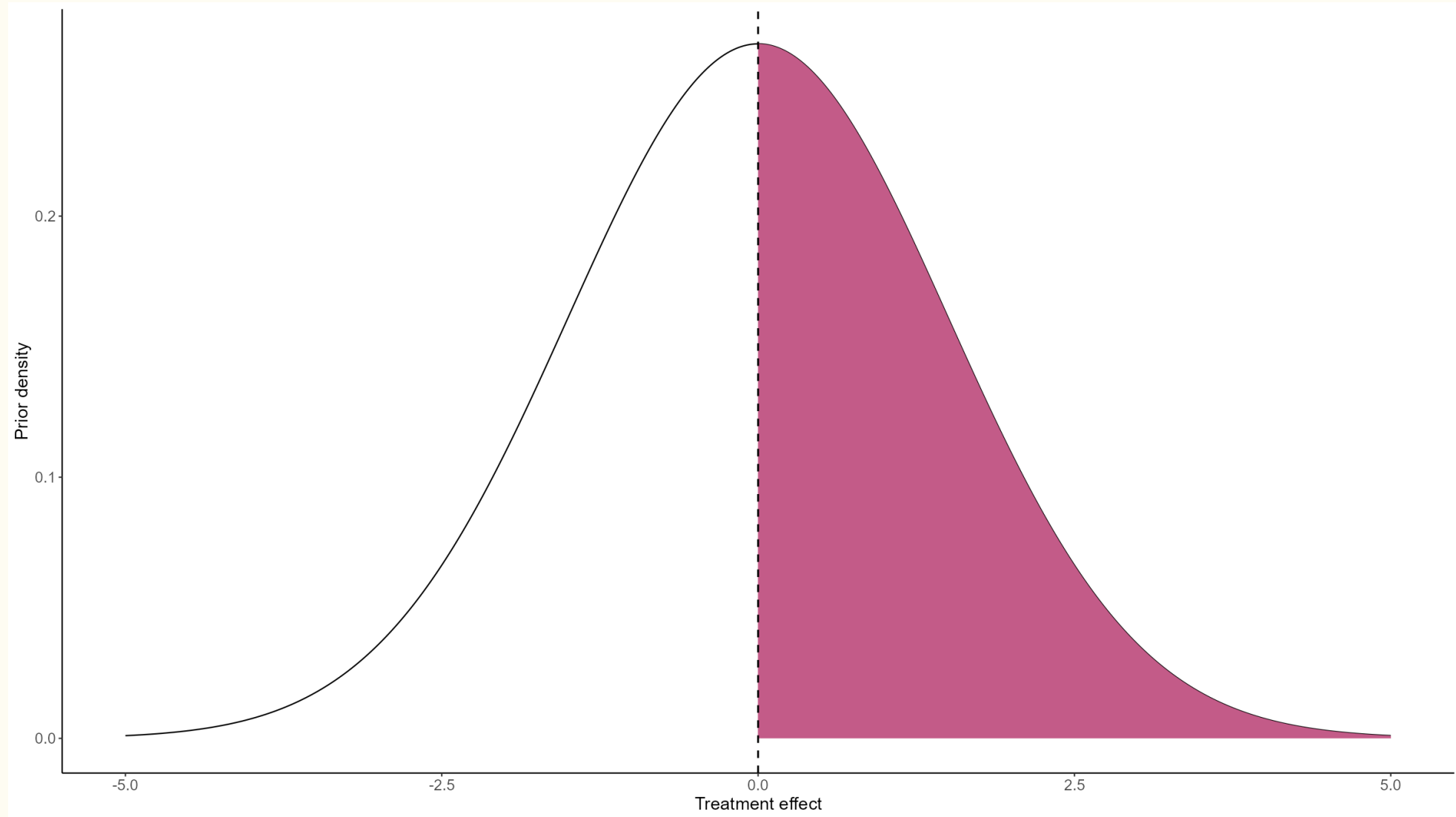
BAYES FACTOR COMPUTATION



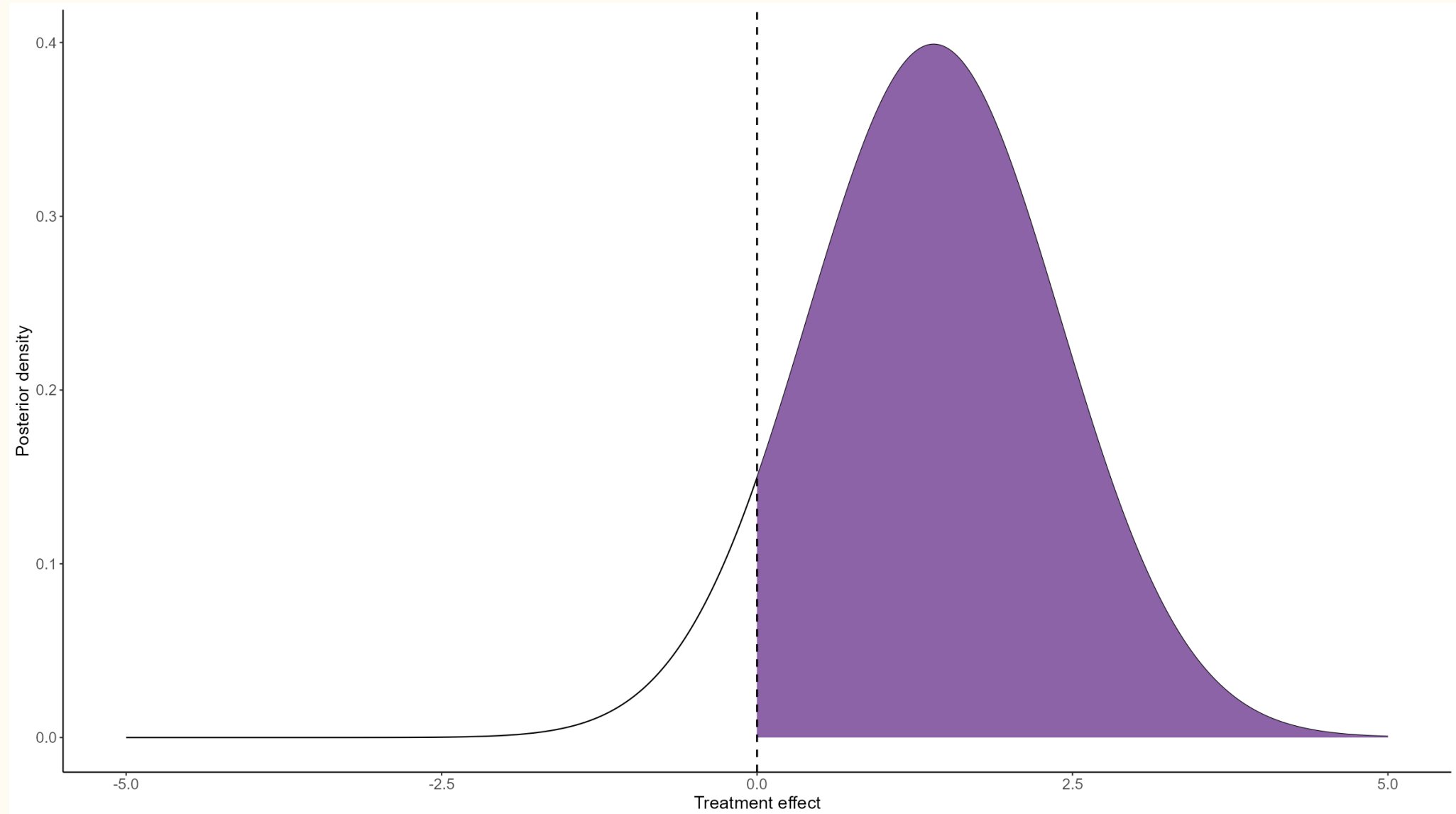
PRIOR DISTRIBUTION - COMPLEXITY

$$H_0 : \mu_C = \mu_T$$

$$H_1 : \mu_C < \mu_T$$



POSTERIOR DISTRIBUTION - FIT



BAYES FACTOR COMPUTATION

$$BF_{1u} = \frac{fit_1}{complexity_1} \longrightarrow BF_{1u} = \frac{0.92}{0.5} = 1.84$$

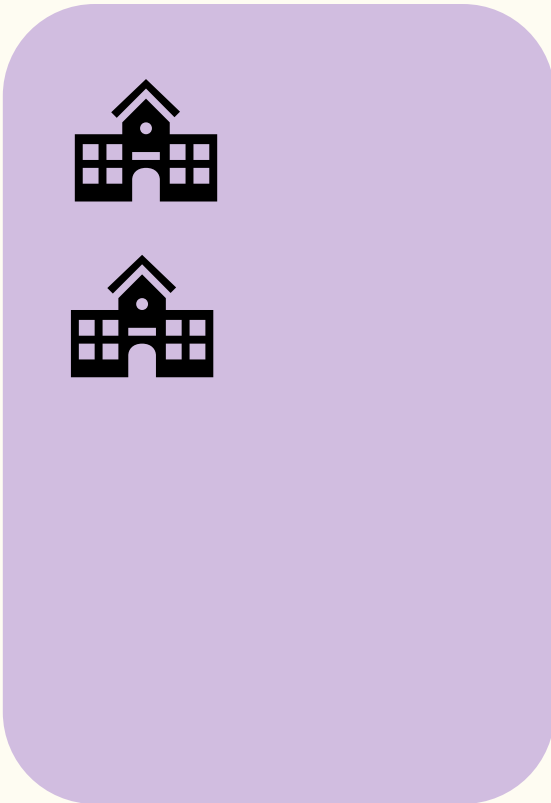
$$BF_{0u} = \frac{fit_0}{complexity_0} \longrightarrow BF_{0u} = \frac{0.15}{0.27} = 0.55$$

$$BF_{10} = \frac{1.84}{0.55} = 3.34$$

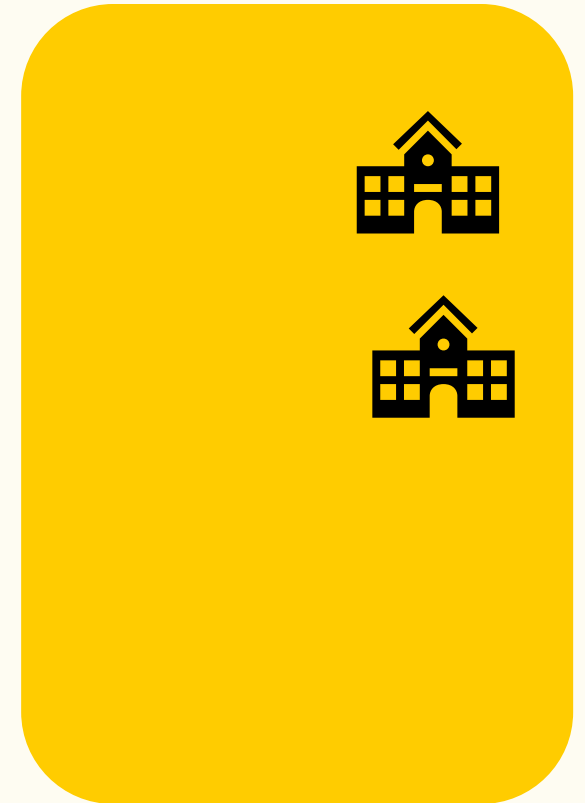
CLUSTER RANDOMISED TRIALS

- Intraclass Correlation Coefficient

Treatment



Control



APPLICATION

BAYESIAN SAMPLE SIZE DETERMINATION

- Pair 1: Null vs. Alternative

$$H_0: \mu_C = \mu_T$$

$$H_1: \mu_C < \mu_T$$

- Pair 2: Informative hypotheses

$$H_1: \mu_C < \mu_T$$

$$H_2: \mu_C > \mu_T$$

- Effect size
- Intraclass correlation coefficient
- Bayes factor threshold
- Bayesian power (η)
- Pair of hypotheses
- Fixed cluster size or number of clusters

Evaluate the efficacy of the FUNBALL injury prevention programme among football players

Treatment condition: FUNBALL programme

Control: FIFA 11+ programme

Randomisation unit: Football teams

ICC 0.05

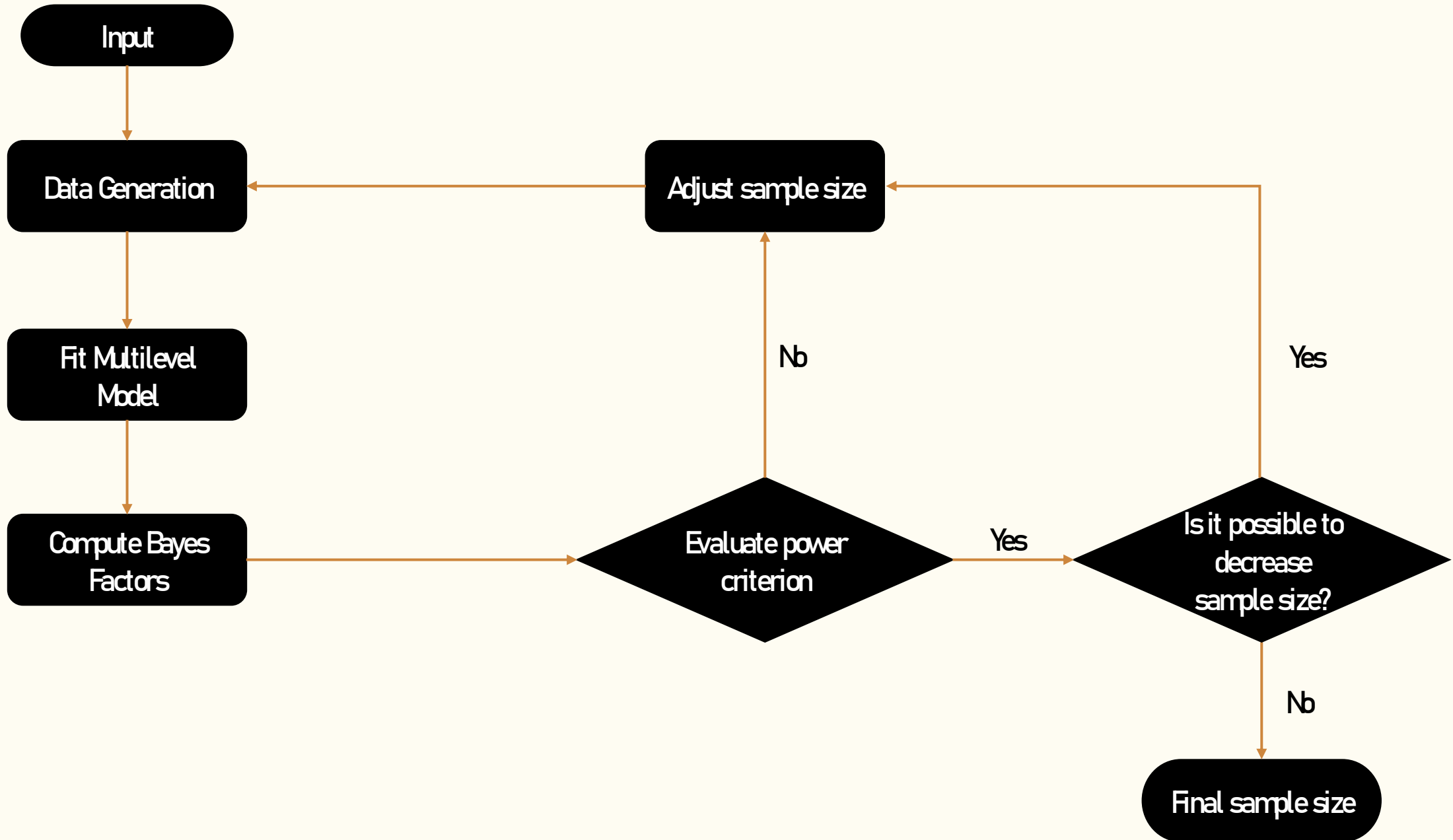
Effect size 0.5

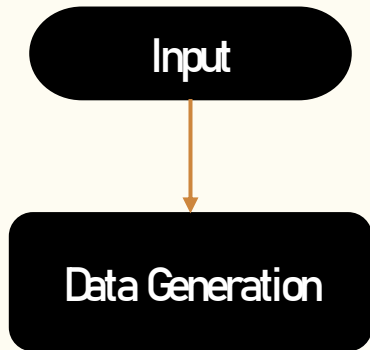
Cluster size: 5 players

Bayes factor threshold: 5

Desired probability: 0.8







N data sets are generated

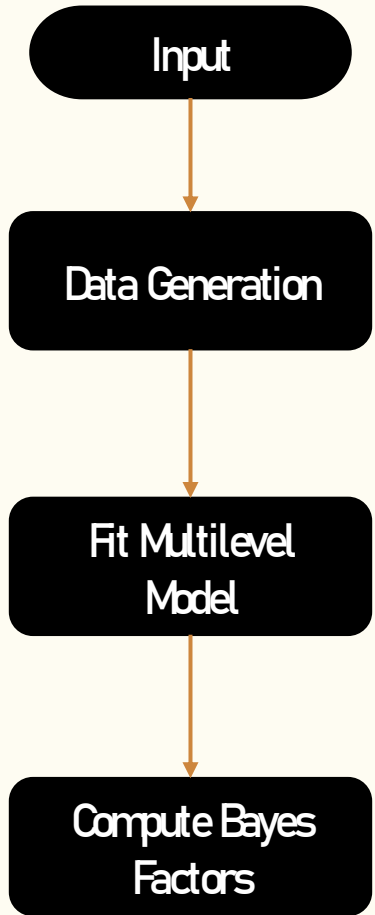
$$Y_{ij} = \beta_{0j} + \beta_{1j}X_j + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{u0}^2 & \sigma_{u01} \\ \sigma_{u10} & \sigma_{u1}^2 \end{pmatrix} \right)$$

$$\epsilon_{ij} \sim N(0, \sigma_e^2)$$



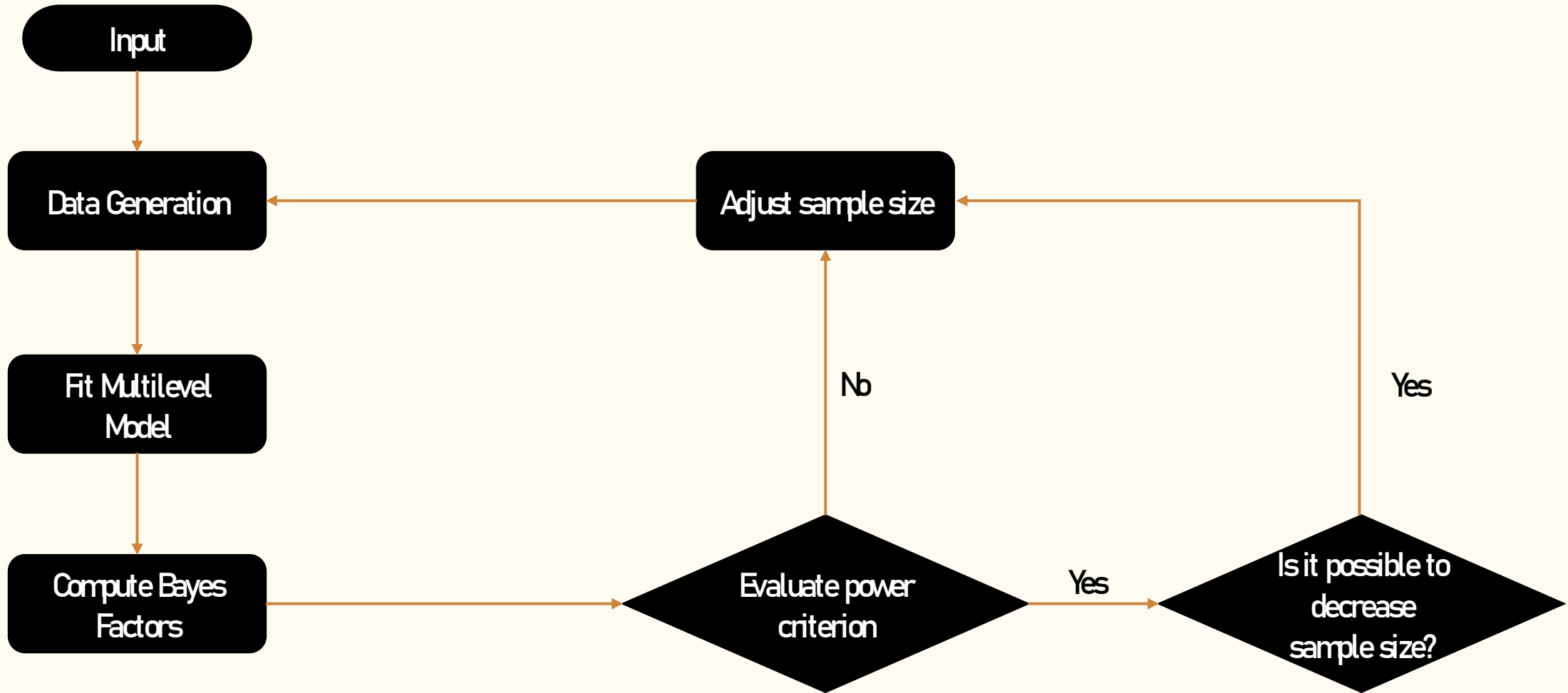
lme4 package
nlme package

$$BF_{12} = \frac{Hypothesis_1}{Hypothesis_2}$$

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_j + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

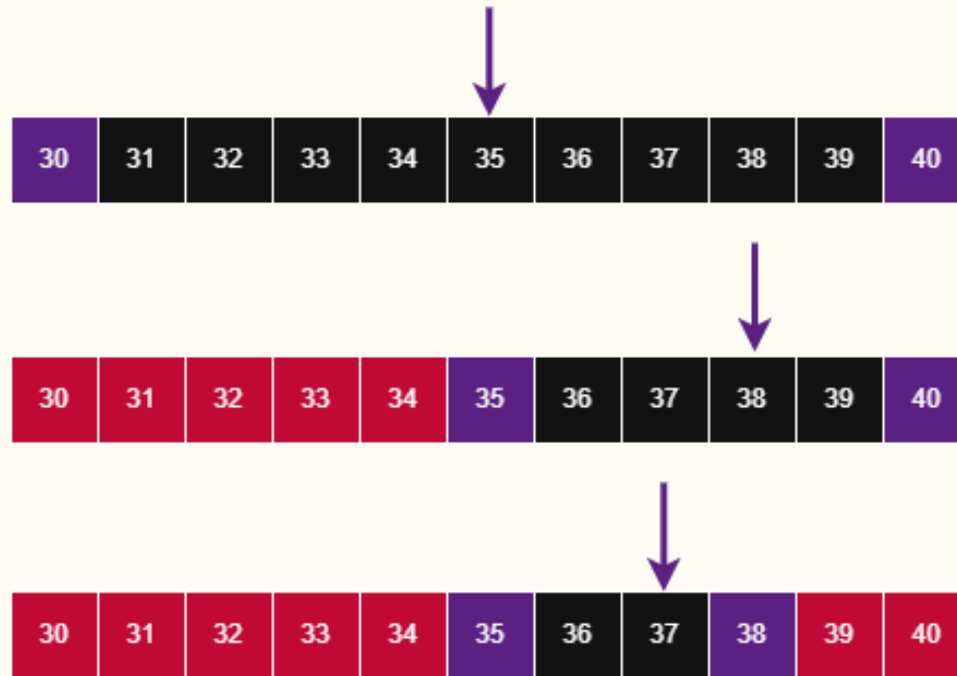
$$\beta_{1j} = \gamma_{10} + u_{1j}$$



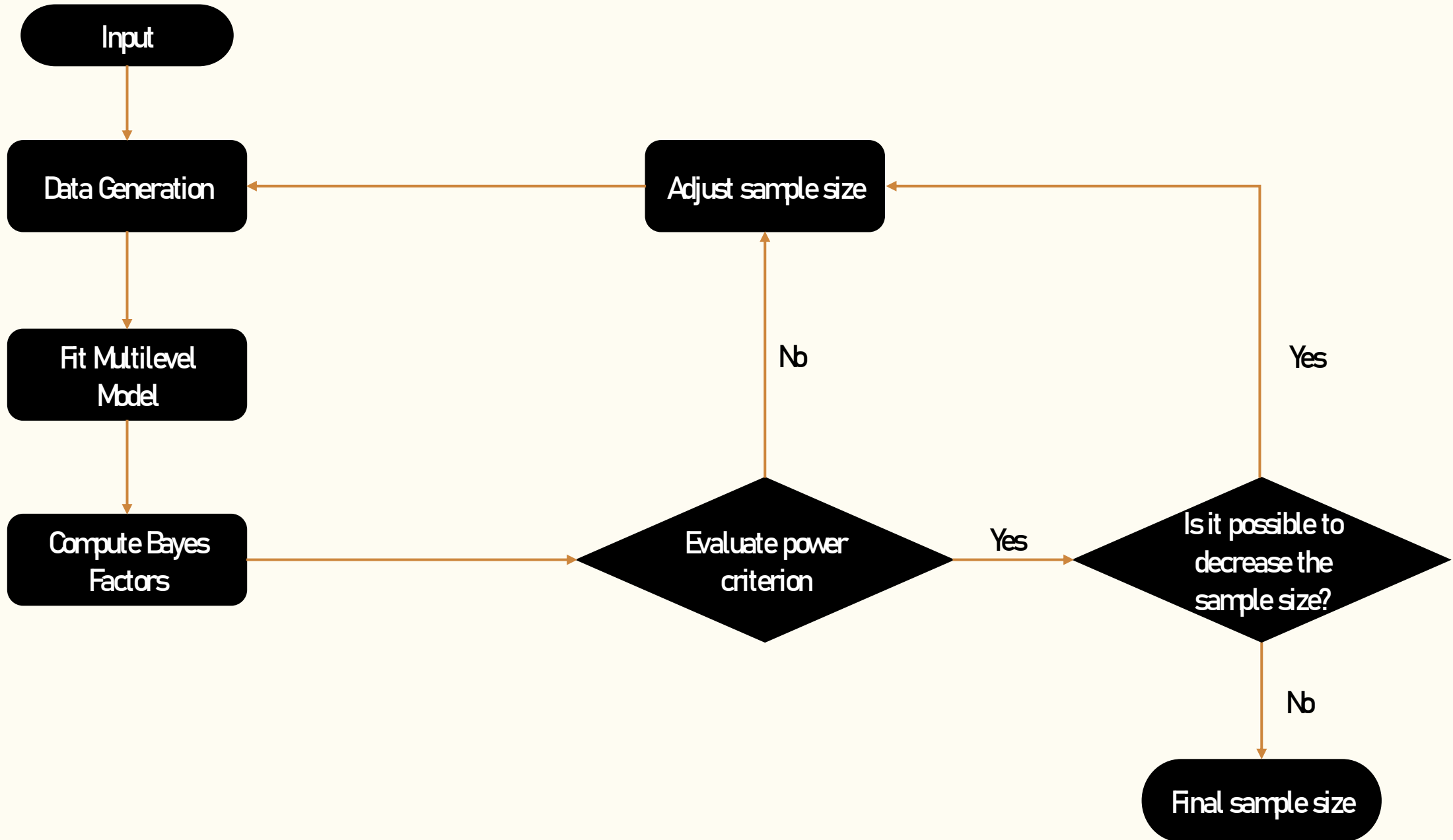
$$P(BF_{01} > BF_{thres} | H_0) \geq \eta$$

$$P(BF_{10} > BF_{thres} | H_1) \geq \eta$$

BINARY SEARCH ALGORITHM



- Adjustment of sample size
- Stops when
 - Lower boundary is close to the middle point
 - The power does not increase anymore.



TWO-ARM CLUSTER RANDOMISED TRIAL

- Sample size increases with
 - High ICC
 - Small effect size
 - Small fixed sample size
 - High Bayes factor threshold
 - Including null hypothesis
- It is easier to reach the power criterion when increasing the number of clusters.
- In some scenarios the power criterion was not met after large cluster size (1000).

Sample Size Determination for Cluster Randomised Trials with Bayes Factor

Results Information

The sample size to be found given the other one is fixed:

☒ Number of clusters

☐ Cluster sizes

Type of hypotheses to compare:

☒ Null vs. Informative

☐ Informative vs. Informative

Cluster sizes:

5

Intraclass correlation:

0.025

Effect size:

0.2

Bayes factor threshold:

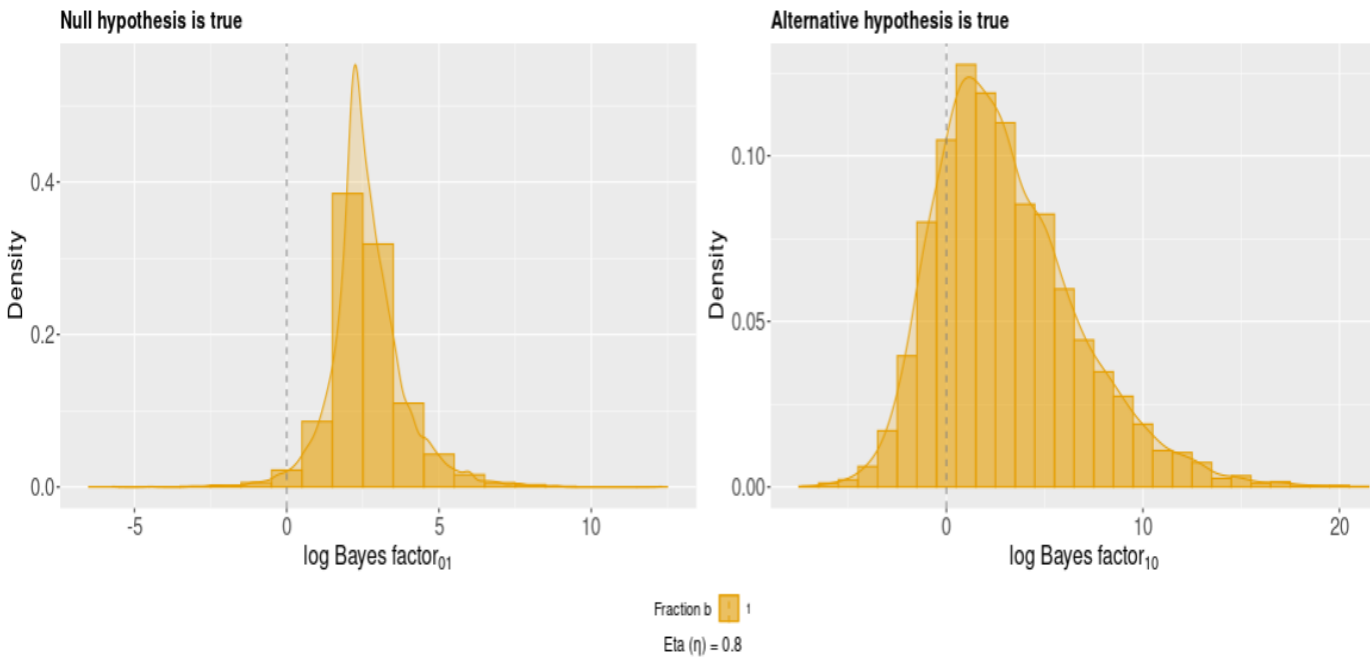
1

Fraction b in plots:

☒ 1

☐ 2

☐ 3



Final Sample Size

b	Number of clusters	P(BF.01 > thresh)	P(BF.10 > thresh)
1.000	126.000	0.982	0.807
2.000	114.000	0.971	0.801
3.000	108.000	0.963	0.803



SAMPLE SIZE DETERMINATION IN MULTIT



MULTIVARIATE CLUSTER RANDOMISED TRIAL

- Effect sizes
- Outcome-specific ICC
- Intersubject between-outcome ICC
- Intrasubject between-outcome ICC
- Fixed cluster size or number of cluster
- Posterior Model Probabilities threshold
- Bayesian power (η)
- Test

$$\mathbf{y}_{ij} = \begin{bmatrix} \gamma_1 \\ \vdots \\ \gamma_p \end{bmatrix} + \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} z_j + \begin{bmatrix} u_{j1} \\ \vdots \\ u_{jp} \end{bmatrix} + \begin{bmatrix} e_{ij1} \\ \vdots \\ e_{ijp} \end{bmatrix}$$

$$P(PMP_m > PMP_{thresh} | H_m) \geq \eta$$

- Intersection-union Test
- Homogeneity of Treatment Effects Test
- Omnibus Test

INTERSECTION-UNIONTEST

In NHST, $H_0: \beta_p = 0$ and is only rejected when $H_1: \beta_p > 0$ for all outcomes p

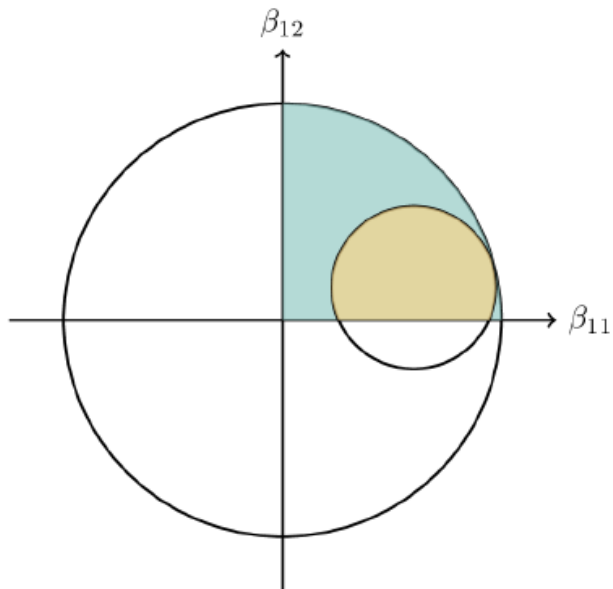
$$H_1: \beta_1 > 0 \text{ \& } \beta_2 > 0$$

$$H_2: \beta_1 > 0 \text{ \& } \beta_2 < 0$$

$$H_3: \beta_1 < 0 \text{ \& } \beta_2 > 0$$

$$H_4: \beta_1 < 0 \text{ \& } \beta_2 < 0$$

The treatment has an effect on both outcomes.



OMNIBUSTEST

In NHST, $H_0: \beta_p = \beta_p$ and is only rejected when at least one β_p is different to other β_p $H_1: \beta_p \neq \beta_p$

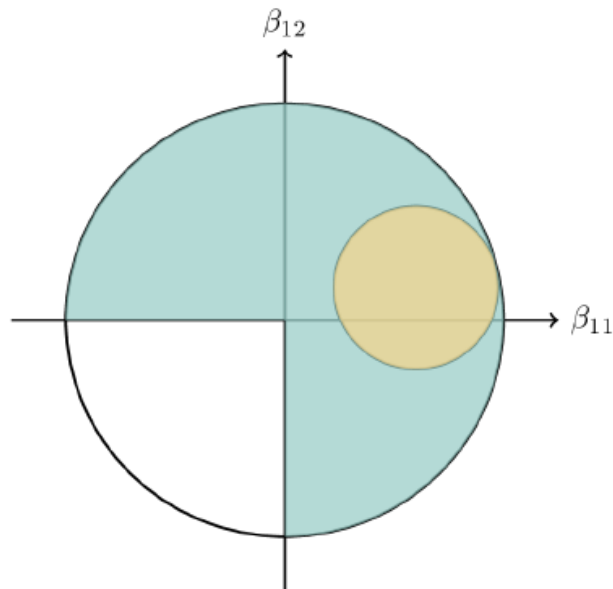
$$H_1: \beta_1 > 0 \text{ \& } \beta_2 > 0$$

$$H_2: \beta_1 > 0 \text{ \& } \beta_2 < 0$$

$$H_3: \beta_1 < 0 \text{ \& } \beta_2 > 0$$

$$H_4: \beta_1 < 0 \text{ \& } \beta_2 < 0$$

The treatment has an effect on at least one of the outcomes.



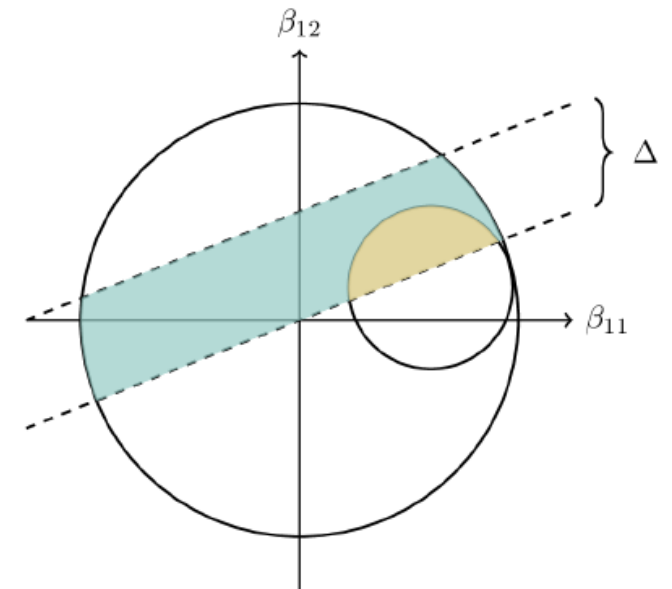
TEST OF HOMOGENEITY OF EFFECT SIZES

In NHST, $H_0: \beta_p = 0_p$ and is only rejected if one β_p is different to 0_p $H_1: \beta_p \neq 0$

$$H_1: \beta_1 - \beta_2 < \Delta$$

$$H_2: \beta_1 - \beta_2 > \Delta$$

The treatment effects are sufficiently similar to be considered equal.



MULTIVARIATE CLUSTER RANDOMISED TRIAL

- The number of clusters increases  when:
 - The outcome-specific ICC increases 
 - The intrasubject between-outcomes ICC increases 
 - The intersubject between-outcomes ICC increases 
 - Small cluster size 
 - Small effect sizes 
 - Posterior model probability threshold increases 

Sample Size Determination for Bayesian Hypothesis Testing in Multivariate Cluster Randomised Trials

Results Information

Type of test

Intersection-union

Cluster size

5

Outcome-specific ICC for outcome 1

0.01

Intersubject between-outcome ICC

0.005

Intrasubject between-outcome ICC

0.2

Posterior Model Probability threshold

0.9

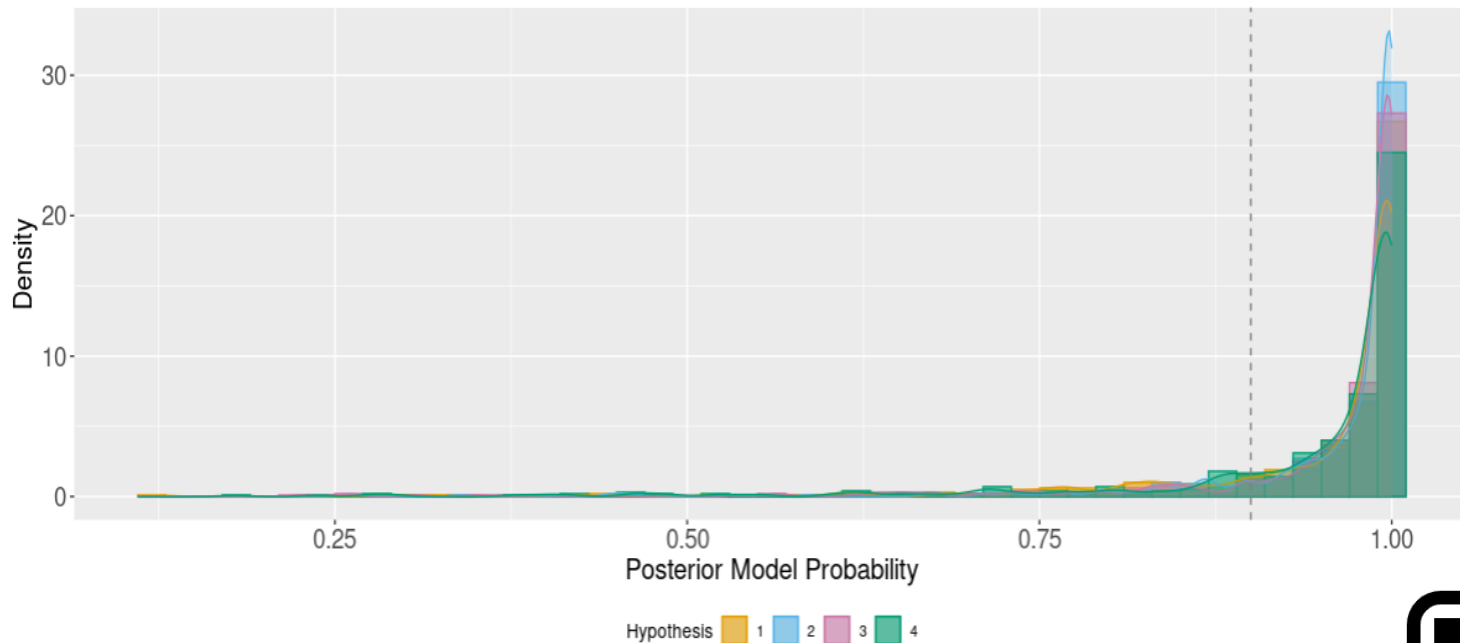
Treatment effect in outcome 1

0.3

Treatment effect in outcome 2

0.5

Posterior Model Probabilities favouring true hypothesis



Final Sample Size

Hypothesis	P(PMP.H > threshold)	Error
H1	0.824	0.055
H2	0.884	0.039
H3	0.868	0.047



CROSSOVER CLUSTER RANDOMISED TRIAL

	Period 1	Period 2
Sequence 1	Control	Treatment
Sequence 2	Treatment	Control

$$y_{ikp} = \mu + \beta_p + X_{kp}\theta + C_k + G_{kp} + e_{ikp}$$

$$C_k \sim N(0, \sigma_c^2), \quad G_{kp} \sim N(0, \sigma_g^2) \quad e_{ikp} \sim N(0, \sigma_e^2)$$

Fixed effect for time period

Treatment effect

Treatment indicator

Cluster-level random effect

Cluster-period random effect

Residual term

CROSSOVER CLUSTER RANDOMISED TRIAL

$$y_{ikp} = \mu + \beta_p + X_{kp}\theta + C_k + G_{kp} + e_{ikp}$$
$$C_k \sim N(0, \sigma_c^2), \quad G_{kp} \sim N(0, \sigma_g^2) \quad e_{ikp} \sim N(0, \sigma_e^2)$$

Fixed effect for time period

Treatment effect

Treatment indicator

Cluster-level random effect

Cluster-period random effect

Residual term

Within-period IOC

$$\rho_{wp} = \frac{\sigma_C^2 + \sigma_G^2}{\sigma_C^2 + \sigma_G^2 + \sigma_e^2}$$

Between-period IOC

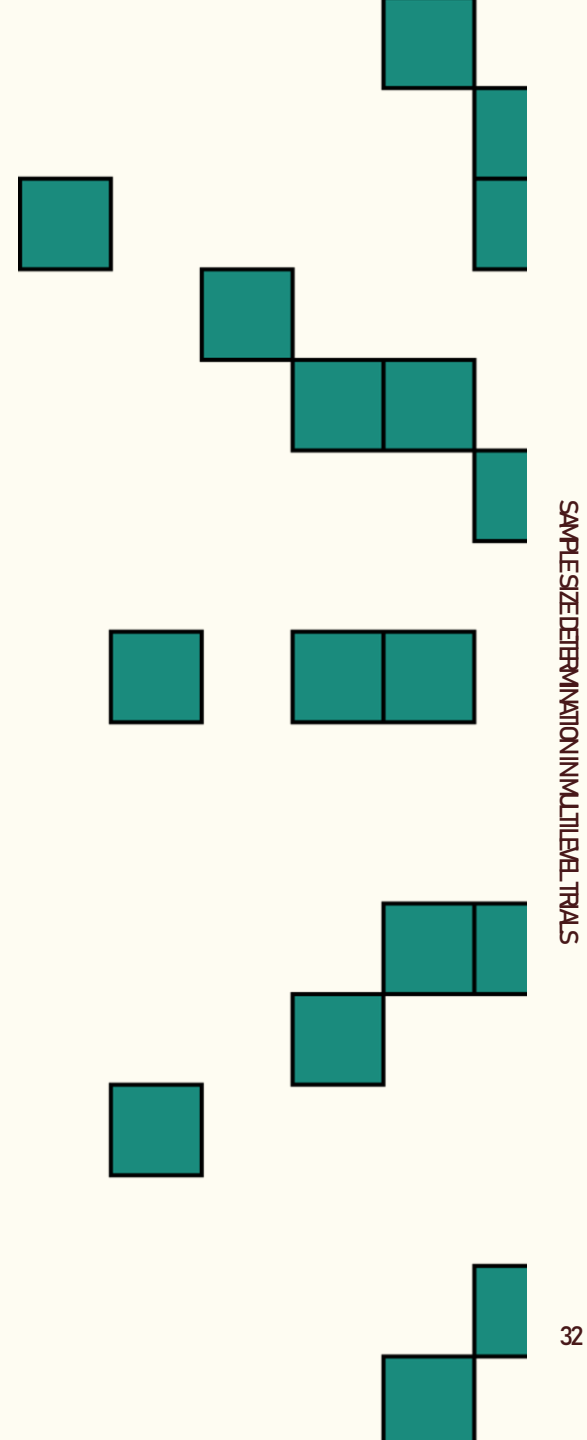
$$\rho_{bp} = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_G^2 + \sigma_e^2}$$

Cluster Autocorrelation Coefficient

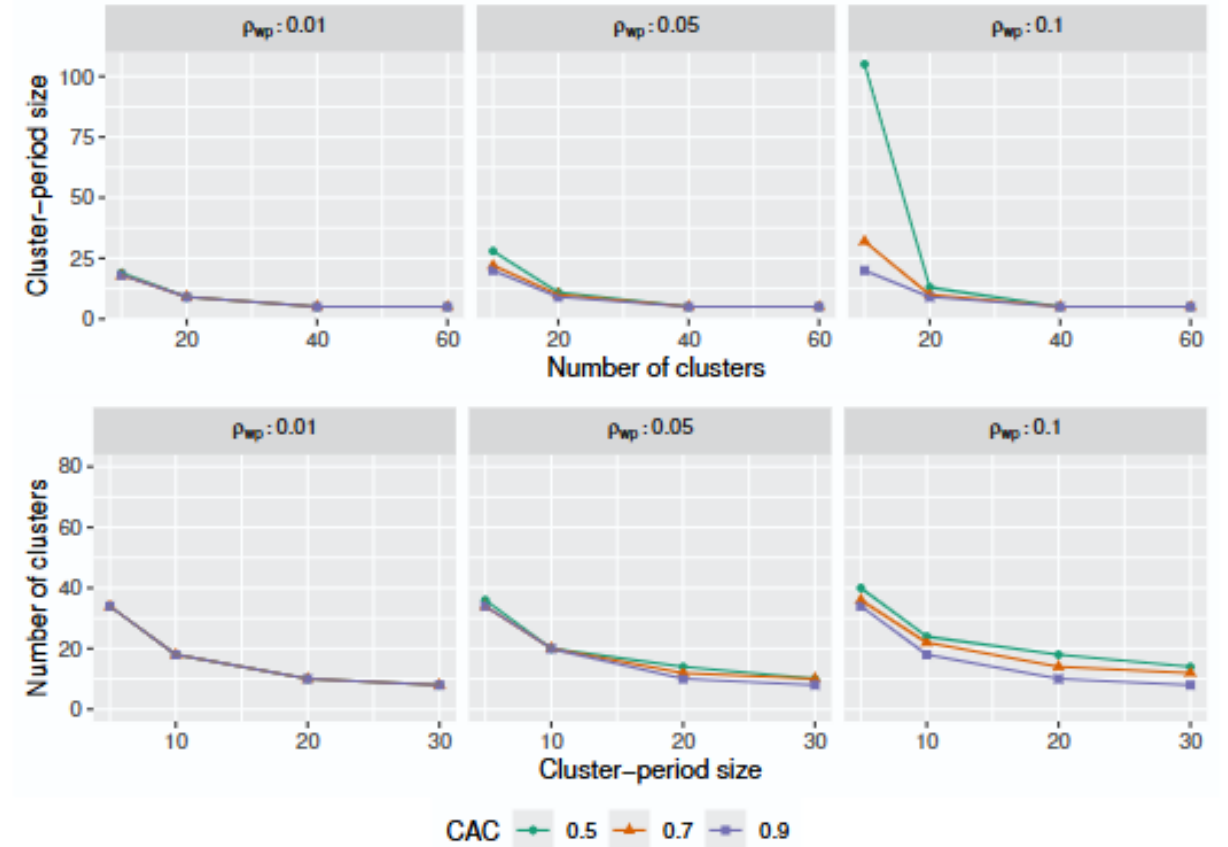
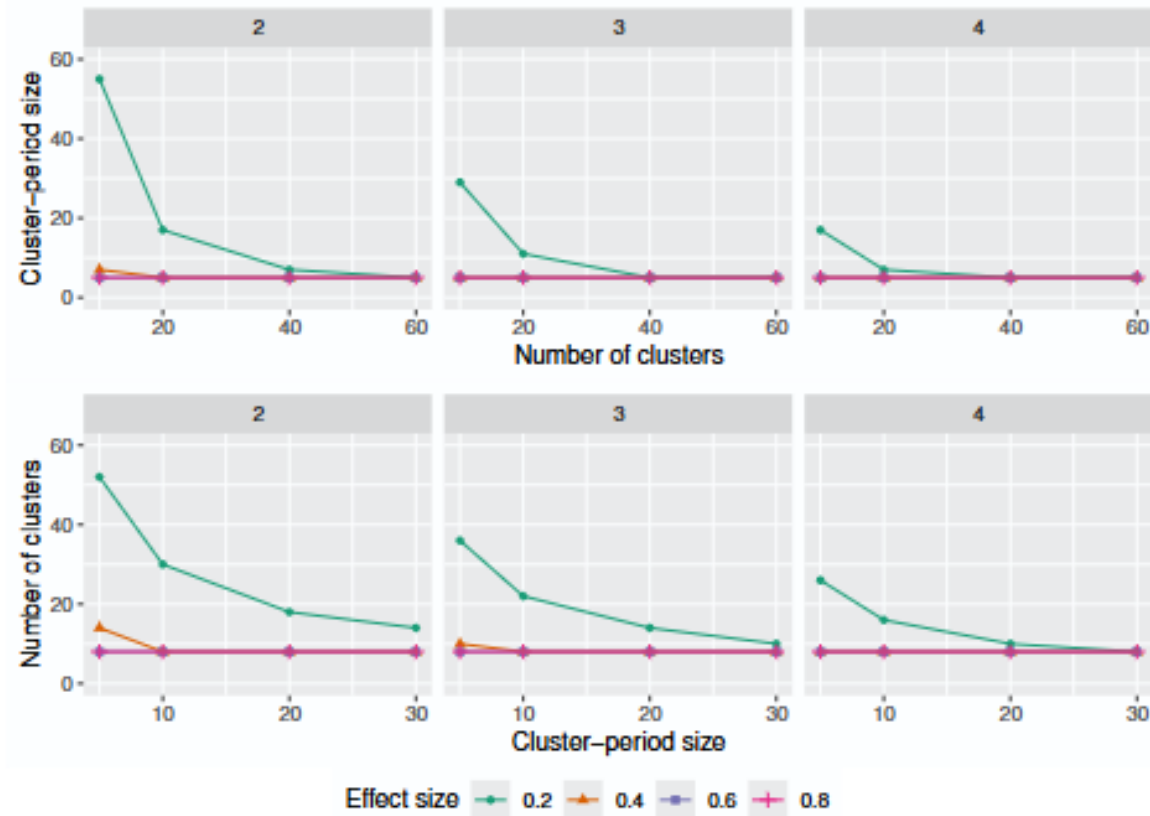
$$CAC = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_G^2}$$

CROSSOVER CLUSTER RANDOMISED TRIAL

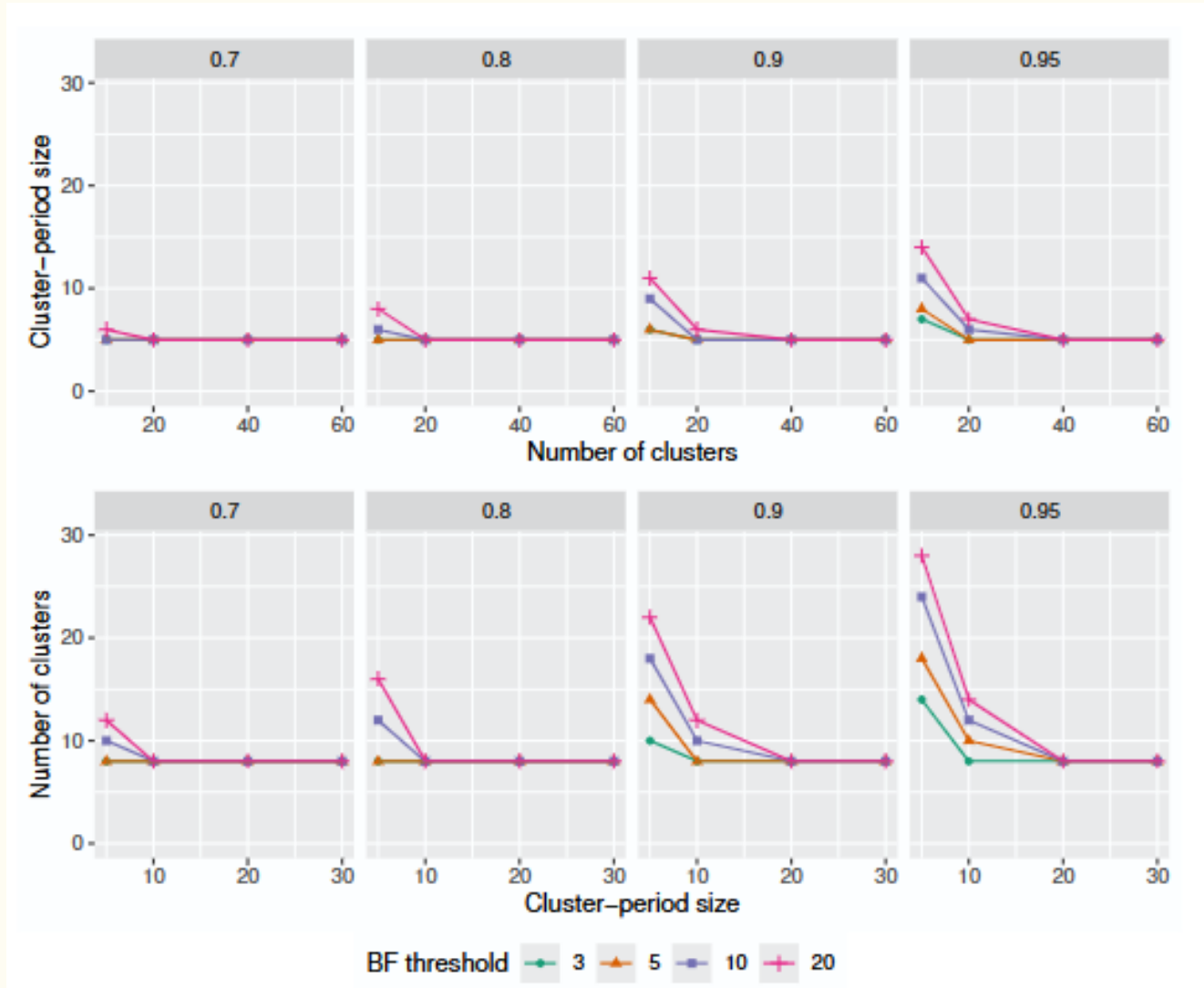
- Effect size
- Within-period ICC
- Cluster Autocorrelation Coefficient
- Bayes factor threshold
- Bayesian power (η)
- Pair of hypotheses
- Fixed cluster-period size or number of clusters



SIMULATION RESULTS



SIMULATION RESULTS



CROSSOVER CLUSTER RANDOMISED TRIAL

- The sample size increases when:
 - Small effect sizes
 - Lower number of periods
 - Small fixed sample size
 - Higher within-period ICC
 - Lower CAC
 - Higher BF threshold
 - Higher probability

LIMITATIONS

Two-arm cluster randomised trials:

Assumption of equal cluster size

Heterogeneity of treatment effect is not considered

Sensitivity to chosen parameters

Multivariate cluster randomised trial:

Assumption of equal cluster size

Heterogeneity of treatment effect is not considered

Sensitivity to chosen parameters

Two outcomes

Crossover cluster randomised trial:

Assumption of equal cluster size and number of clusters

Heterogeneity of treatment effect is not considered

Sensitivity to chosen parameters

Limited to a design in which there is a crossover every period

Limited to two sequences

CONCLUSIONS

BayesSSD is useful tool for researchers designing a (crossover) cluster randomized trial.

The methodology represents an alternative to researchers that want to use Bayesian hypothesis testing but have limited knowledge about it.

Thank you for your attention!